

Jyothir Anurag Vasu Chinnaboina

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Education

University of Wisconsin–Madison – *B.S. in Computer Science and Data Science*

Madison, WI

Awards: Dean's List (4 semesters)

Expected May 2026

Coursework: Operating Systems, Algorithms, AI/ML, Statistics, Database Management, Data Structures

GPA: 3.7/4.0

Projects

F1 Race Strategist (RL) | *Python, FastAPI, Pydantic 2, Docker, PyTorch, Hugging Face*

Present

Meta PyTorch OpenEnv Hackathon – Long-Horizon Planning Track · fl.chinnaboina.com

- Trained Qwen3-4B (SFT + GRPO, TRL/LoRA) to 0.618 (48% gain), matching expert on weather scenarios (0.965 vs 0.950) via a 6-D Python reward.
- Shipped an RL environment server (FastAPI/Pydantic 2, Docker) used by 4 judges and 22 competing teams via a Gradio UI on Hugging Face Spaces.
- Modeled 22 F1 tracks with tire/fuel/weather state across 5 endpoints with concurrent rollouts, gated by agent inspection queries.

ApplyOps | *Python, Claude Agent SDK, ChromaDB, OpenAI Embeddings, asyncio, SQLite*

Present

Personal pipeline infrastructure · github.com/Anurax1321/ApplyOps

- Designed and built a multi-agent tailoring pipeline (Claude, ChromaDB) saving ~80K tokens/app across 250+ apps via Writer/Critic loops.
- Crafted an asyncio daemon (5 vector collections, sub-second from 30s cold) auto-injecting bullets and memory into every Claude conversation.
- Routed 250+ JDs through a 4-tier similarity router (Seed/Reference/Standard/Generic) auto-reusing prior resumes to save 20-78K tokens/JD.
- Validated 3 concurrent runs to zero SQLite errors via WAL + rebuild locks; 26 pytest tests across retrieval, cache, and daemon fallback.

Office Hours Queue | *React/TypeScript, Go/Gin, PostgreSQL, Redis, Docker*

Sep 2025 – Present

Co-Founder · *Advisor: Prof. Louis Oliphant (CS)* · ohq.cs.wisc.edu

- Built an agentic code-analysis pipeline (MCP, 9 tools) sandboxed in Docker (512MB cap, non-root, 60s timeout) across 2 CS courses.
- Built a RAG assistant (FastAPI, ChromaDB, GPT-4o mini, Qwen3-Embedding-8B) with semantic search across 4 sources, ~2s lookup.
- Reduced wait 1–3hr to 15min for 500+ students with SSE + Redis pub/sub; migrating Flask to Go/Gin for 10x concurrent SSE.

Experience

Software Developer Intern (*Part-time*) – *Vijay Rekha Life Sciences*

Aug 2025 – Present

Sole Developer · vijayrekha.chinnaboina.com

- Launched a neonatal lab system (React/TS, FastAPI, PostgreSQL) processing 300+ samples/day with automated validation and flagging for 4 techs.
- Accelerated clinical report validation by 85% with a batch pipeline (pdfplumber, regex) cross-validating patient data from 100+ PDFs.
- Automated 3+ hours/day of manual work via a Python pipeline (pandas, xlsxwriter) transforming mass spectrometry data for 40+ compounds.

Mainframe Intern (*Part-time*) – *Wisconsin Department of Administration*

Jun 2025 – Present

- Reduced manual triage time by 90% with a Python log parser surfacing remediation status and follow-up priority for 5+ mainframe teams.
- Deployed an IaC platform (Ansible, Terraform) adopted department-wide, streamlining 15+ batch job workflows across state systems.

Computer Science Tutor – *Undergraduate Learning Center, UW–Madison*

Sep 2024 – Aug 2025

- Mentored 600+ students across 8 CS courses and evaluated 160+ tutor candidates across 3 hiring cycles, designing technical interview questions.

ERP Integration Intern – *United States Pharmacopeia*

Jun – Aug 2024

- Improved data retrieval by 70% with an Oracle ERP tool (Python, oracledb, pandas) using SQL joins on 5 tables to surface order hold trends.

Skills

Languages: TypeScript, JavaScript, Python, Go, Java, C/C++, SQL, Bash, HTML/CSS

Frameworks & Tools: React, Gin, Flask, FastAPI, Pydantic, Uvicorn, Node.js/Express, Docker, Docker Compose, Terraform, Git, GitHub Actions

Data & ML: PyTorch, Hugging Face, OpenEnv, scikit-learn, ChromaDB, OpenAI API, Claude Agent SDK, pandas, NumPy, Matplotlib

Databases & Cloud: PostgreSQL, Redis, Oracle DB, MySQL, SQLite, GCP, Linux

Other: REST APIs, SSE, Gradio, asyncio, OAuth2, JWT, CI/CD, Pytest